

TEMP, CO₂, O₂, humidity controlled by 4 systems.
for modular design

TEMP: • Temperature module connects to objective

• temp module also heats up the
circulating gas mixture

→ Heating insert

→ Control sensor

→ Incubator

→ Humidity device

CO₂: CO₂ module changes CO₂ concentration in Gas mixture

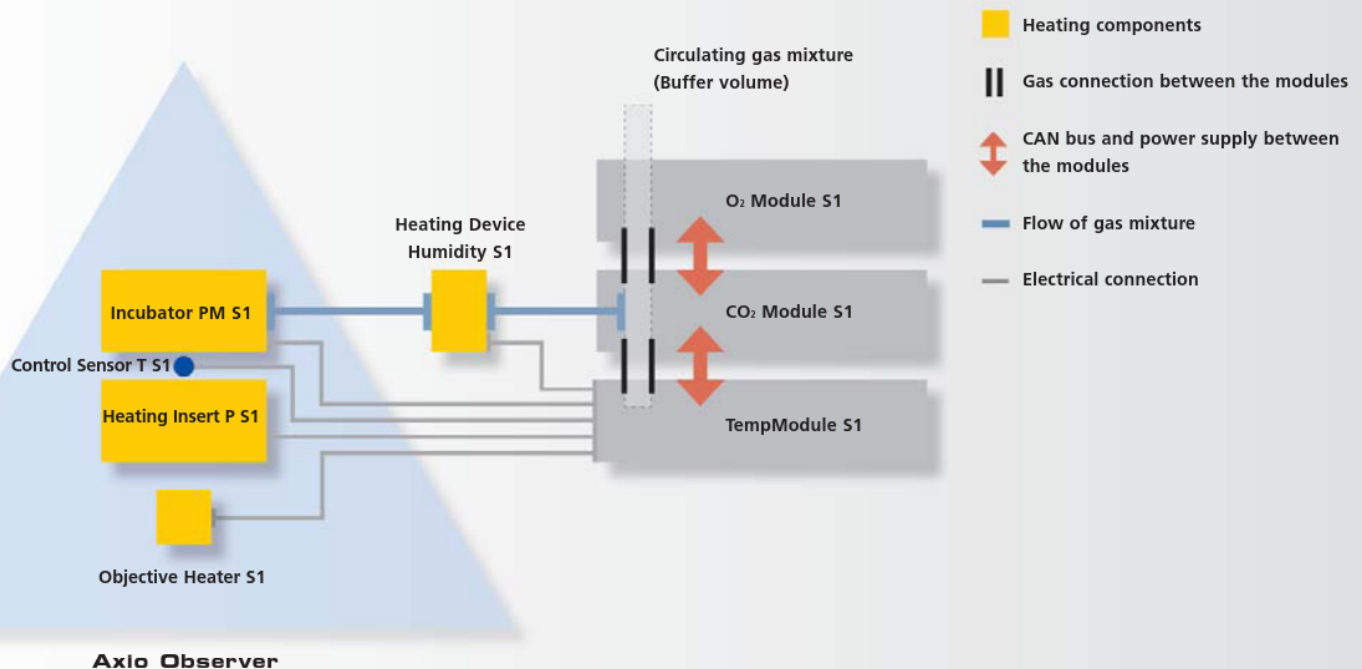
O₂: O₂ module changes O₂ concentration in Gas mixture

Humidity: separate device that can be heated → Takes in dry Gas

→ Adds humidity

→ Puffs resulting mixture
to Incubator

The system design - shown using the example of the Incubator PM S1



Zeiss Software can record all data from Temp, CO_2 , Humidity apparatus with the microscope recordings to allow for better correlation between effects

TEMP:
TEMP module supplies power and control signals to other modules.

Internal resolution: $0.01^\circ C$

can be connected to control sensor TSI to directly measure culture temp

Plus 4 independent heating channels (objective, insert, incubator humidity)

Heating Range: Ambient to $60.0^\circ C$
(Recommended Ambient to $45^\circ C$)

Don't think Control sensor Valid for BSL3!

↑
depends on it viable to leave in and clean

CO₂:

Built in CO₂ sensor, No direct measurement (O₂ sensitive?)

Internal resolution 0.01 %.

Setpoint range 0.0 % $\Rightarrow$ 8.0 %.

O₂ module:

Built in O₂ sensor (why mention extremely durable?)

Internal res: 0.01%.

Setpoint Range: 0.0 - 21.0 %.

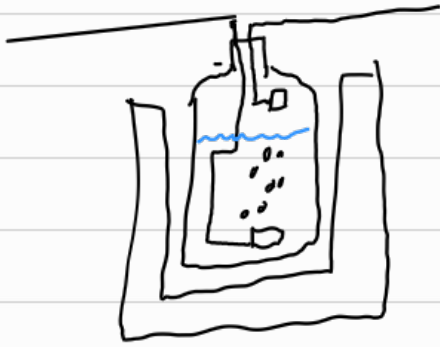
Zeis Incubator :

- Perfect Seal $\rightarrow$ How perfect $\rightarrow$ Not apparently
 - Gas go in, but how gas go out? $\rightarrow$ Ventral out the bottom
 - $\hookrightarrow$ b evenly distributed, internal openings ensure that the supply of gas is homogeneously distributed
 - Notches for tubes and cables $\rightarrow$ Not Ideal in BSL3!
 - External dimen (L x W x H mm) 132 x 205 x 18
 - Observation area (L x W mm) 120 x 80
 - Glass heated directly with embedded elements
- Heating Insert P Lab-TekSM S1 (use this since compatible with slides)
- 4 leveling screws (46 x 21 mm) — for Lab-Tek only
 - Oval opening in base (32 x 33 mm) for high aperture objective
 - Port for perfusion tubes on both left and right side.
 - 2 ports for perfusion, control sensor T S1 at front
 - Heated grooves for perfusion tubes, for temp matching
 - dimensions (L x W x H mm) 160 x 135 x 22

Heating device Humidity S1

- bottle sits in heating cylinder
- Gas bubbled through water to humidify it
 - (Not BSL3 safe?)
hmm depends on filters I guess

Temp
and CO₂



→ Incubator

Overview : Carl Zeiss / PeCon commercial solution

Incubator System broken up into 2 subsystems.

• 1: Incubator : 2 parts $\Rightarrow$ Incubator Lid
 $\Rightarrow$ Heating insert

- Both parts can be heated by Temp module
- heating insert accepts growth media $\Rightarrow$ has opening at bottom
 $\hookrightarrow$ Incubator Lid "seals" the unit

the Incubator does not form a complete seal as it is intended for the Gas mixture to enter from the tube, then flow over the media, then get exhausted out of openings in the seal. Not ideal for BSL3

The lid and Insert each have a temp sensor for temp control

• 2: Control : 1. temp module : heats up each part and can individually control each

- heats up Gas mixture

- heats up humidifier

2. CO_2 module : controls release of CO_2 into Gas mixture to maintain CO_2 concentration for pH control

3. humidity module: • bubbles Gas mixture through heated sterile water to increase Relative humidity of mixture to 100%.

